



Autonomous car and self-driving concept

standards for different vehicle categories (two-wheeler, three-wheeler, four-wheeler, and heavy electric vehicle). These are 'open' and freely available, motivating multiple startups to come up with new classes of EVs.

Startups looking to launch their products in the market can take help of *gyaniki*, a knowledge portal based on past experiences of automotive designers, where a lot of information is given on EVs, autonomous vehicles, and related component manufacturers.

Intelligent vehicle challenges

Recounting his PhD days in the UK back in 2010, Dr Rahul Kala, now an assistant professor at the Centre of Intelligent Robotics - IIIT Allahabad, says that one day while walking from his residence to workplace, he witnessed chaos in a nearby street. Due to some reason, one of the lanes in a two-lane British road was blocked. Yet the drivers cleverly managed to get past each other without crashing. This made him think... can self-driving cars do the same?

Considering this scenario, it can be interpreted that autonomous machines fail to produce the same level of perception at present. Humans generally deal with machines through non-verbal communication, which is unlike a verbal human-to-human conversation. Vehicle intelligence is not based on the number of miles tested but on gauging the level of human intelligence (or ignorance) around testing. And the only way of knowing it is by analysing the behaviour of people with poor driving skills around those with good

driving skills. This test can enable an intelligent vehicle to match up to the chaos that exists on roads.

Another thing to note is the uncertainty of local weather conditions. All in all, it is a question of dealing with the unpredictability of humans and nature. If the situations are handled well, then it can safely be said that we have reached the levels of testing in autonomy.

"Observation is the key. The more you observe, the more key points you get for validating the test cases," says Naresh Neelakantan, senior architect - system software at Sasken Technologies.

Cloud computing challenges

Autonomous cars together utilise multiple sensor inputs to collect enormous data through vision processing and object detection to prevent road accidents. However, the biggest challenge is efficiently processing and delivering high data volumes on the existing communication network. Even if 5G is available everywhere, it cannot provide the required low-enough latency. A delay of even a millisecond could result in a fatal collision.

"What happens if the data connectivity fails while on the road? We cannot rely on the cloud infrastructure for every process," says Arqum Ahmad, CEO, Antrax Labs and AI scientist at Leonardo, Italy (Military Aircraft Division). "We need an autonomous car that can perform object detection and relevant decision making without any delay. This

challenge is not just limited to handling the ever exploding data but also ensuring security and reliability." And this is where edge computing plays a huge role. Edge computing offers low latency and faster decision making on the local site. Keeping this in mind, the Automotive Edge Computing Consortium (AECC)—a group of tech giants like Google and Tesla, universities, and start-ups—has been established to focus on convergence between vehicles and the computing ecosystem.

NVIDIA has already launched a product called Drive DGX, which is an edge node having faster GPUs and to which multiple sensors can be connected. One can use it as a testbed and train a model on the autonomous car itself.

Edge computing also plays an important role in EVs. With the help of predictive maintenance, anything going wrong with the battery can be determined (overheating is a big issue). And not just prediction, edge computing also facilitates optimisation of the charging process.

This is not to undermine the importance of cloud computing. Mechanisms such as machine-to-machine communication, smart traffic lights, and other smart city infrastructures rely on cloud computing for effective network connection.

Naresh from Sasken agrees and adds, "Unless you have nearby communication infrastructure and bandwidth, the realisation becomes overhead. It is similar to a clogged entry and exit to huge apartments or office spaces."

Securing IoT for test

While there are different concepts about connected cars, it is simply based on four kinds of bi-directional communication with Internet Protocols (IPs): vehicle to users, vehicle to cloud, vehicle to infrastructure, and vehicle to vehicle.

From a validation standpoint, security concerning cloud, edge, and embedded design, and user acceptance with respect to features play a major role. In absence of a testing infrastructure, it would be unfeasible to test driverless cars. And if we test in the lab, then how would the car enter market? Therefore, we need to define the regulation, security levels, and